

Lab 6: Exam 01 Review — Comprehensive Study Session

BIOL-1

Name: _____ **Date:** _____

Overview

This lab session is dedicated to reviewing Modules 1-6 in preparation for Exam 01. It is a paper-based, collaborative study session. You will work in groups to complete review packets, concept maps, and practice questions covering the foundational concepts of biology.

Learning Objectives

By the end of this lab, you will be able to:

1. Connect concepts across Modules 1-6 (Chemistry → Molecules → Cells → Membranes → Metabolism).
2. Identify areas of strength and weakness in your understanding.
3. Practice applying knowledge to new scenarios (similar to exam questions).
4. Collaborate with peers to explain and clarify complex topics.

Materials

- Review Packet (provided)
- Whiteboards and markers
- Textbook/Notes
- Concept Map templates

Activity 1: The Big Picture Concept Map

On a large whiteboard or paper, create a concept map connecting the following terms:

- Atom
- Molecule (Carbohydrate, Lipid, Protein, Nucleic Acid)

- Cell (Prokaryotic, Eukaryotic)
- Organelle (Nucleus, Mitochondria, Chloroplast)
- Membrane (Phospholipid Bilayer)
- Transport (Diffusion, Osmosis, Active Transport)
- Energy (ATP, Enzyme, Metabolism)

Draw arrows connecting related terms and write a verb on the arrow explaining the relationship.

Activity 2: Module Review Stations

Rotate through stations focusing on specific modules:

- **Station 1:** Chemistry & Molecules (Bonding, pH, Macromolecules)
- **Station 2:** Cells & Membranes (Organelle function, Transport types)
- **Station 3:** Metabolism & Enzymes (Energy, Activation Energy, Enzyme regulation)

Activity 3: Practice Exam Questions

Work through specific practice questions from the review packet. Discuss your answers with your group. If you disagree, explain your reasoning until you reach a consensus.

Conclusion

Use the remaining time to ask your instructor specific questions found during the review.